

Assignment 3

Due date: Mar 24th, 2026 before midnight

Problem 1. If X is exponential with rate λ , show that $Y = \lfloor X \rfloor + 1$ is geometric with parameter $p = 1 - e^{-\lambda}$, where $\lfloor X \rfloor$ is the largest integer less than or equal to X .

Solution. We show that Y is a geometric distribution by finding its probability mass function (pmf). Let n be any positive integer, then

$$\begin{aligned} P(Y = n) &= P(\lfloor X \rfloor + 1 = n) \\ &= P(\lfloor X \rfloor = n - 1) \\ &= P(n - 1 < X < n) \\ &= e^{-\lambda(n-1)} - e^{-\lambda n} \\ &= (e^{-\lambda})^{n-1} (1 - e^{-\lambda}) \\ &= (1 - p)^{n-1} p, \end{aligned}$$

which shows that Y is a geometric random variable with parameter $p = 1 - e^{-\lambda}$.

Problem 2. Events occur according to a Poisson process with rate λ . Each time an event occurs, we must decide whether or not to stop, with our objective being to stop at the last event to occur prior to some specified time T , where $T > 1/\lambda$. That is, if an event occurs at time t , $0 \leq t \leq T$, and we decide to stop, then we win if there are no additional events by time T , and we lose otherwise. If we do not stop when an event occurs and no additional events occur by time T , then we lose. Also, if no events occur by time T , then we lose. Consider the strategy that stops at the first event to occur after some fixed time s , $0 \leq s \leq T$.

- (a) Using this strategy, what is the probability of winning?
- (b) What value of s maximizes the probability of winning?
- (c) Show that one's probability of winning when using the preceding strategy with the value of s specified in part (b) is $1/e$.

Solution.

- (a) We will win using this strategy if there is exactly one event occurs within $(s, T]$. Hence,

$$P(\text{probability of winning}) = P(N(T) - N(s) = 1) = \lambda(T - s)e^{-\lambda(T-s)}$$

- (b) Differentiating the expression in part (a) and then setting it equal to 0 gives

$$-\lambda e^{-\lambda(T-s)} + \lambda(T - s)e^{-\lambda(T-s)}\lambda = 0$$

Solving the equation for s gives

$$s = T - \frac{1}{\lambda}$$

- (c) Since $s = T - 1/\lambda$, we have $\lambda(T - s) = 1$ and thus,

$$P(N(T) - N(s) = 1) = \lambda(T - s)e^{-\lambda(T-s)} = e^{-1}$$

Problem 3. A new malware is spreading on the Internet! Our goal is to estimate its spread/damage by time t , assuming it starts at time 0. Assume that internet hosts get infected by this malware according to a Poisson process with rate λ , where λ is **unknown**. Thrasyvoulos installs a Honeypot security system to detect whether hosts are infected. Unfortunately, there is a **lag time** between when a computer is infected by the malware and the Honeypot detects the damage. Assume that the lag time is distributed $\text{Exp}(\mu)$. Suppose that Thrasyvoulos's Honeypot system has detected $N_1(t)$ infected hosts by time t . Thrasyvoulos worries that, because of the lag, the number of infected hosts is actually much higher than $N_1(t)$. The goal of the problem is to understand how many additional hosts, $N_2(t)$, are expected to also be infected at time t .

- Suppose that an infection happens at time s , where $0 < s < t$. What is the probability that the infection is detected at time t ?
- Consider an arbitrary infection that happens before time t . What is the (unconditional) probability p that the infection is detected by the Honeypot by time t ?
- How can we use this knowledge to estimate λ ? (hint: consider using Poisson splitting)
- Use your estimate of λ to determine the expected value of $N_2(t)$.

Solution.

- Given infection at time s , the lag time $L \sim \text{Exp}(\mu)$. The infection is detected by time t if $L \leq t - s$. Thus:

$$P(\text{detected by } t \mid \text{infected at } s) = P(L \leq t - s) = 1 - e^{-\mu(t-s)}$$

- For a Poisson process, given that an infection occurs before time t , its occurrence time S is uniformly distributed on $(0, t)$. Thus:

$$\begin{aligned} p &= E[P(\text{detected by } t \mid S)] \\ &= \frac{1}{t} \int_0^t (1 - e^{-\mu(t-s)}) ds \\ &= \frac{1}{t} \left[t - \int_0^t e^{-\mu(t-s)} ds \right] \\ &= \frac{1}{t} \left[t - \int_0^t e^{-\mu u} du \right] \quad (\text{let } u = t - s). \\ &= \frac{1}{t} \left[t - \left(-\frac{1}{\mu} e^{-\mu u} \Big|_0^t \right) \right] \\ &= \frac{1}{t} \left[t + \frac{1}{\mu} (e^{-\mu t} - 1) \right] \\ &= 1 - \frac{1 - e^{-\mu t}}{\mu t} \end{aligned}$$

- By Poisson splitting, the detected infections $N_1(t)$ form a thinned Poisson process with rate λp . Thus $N_1(t) \sim \text{Poisson}(\lambda p t)$. Given an observation $N_1(t) = n_1$, we can estimate λ by:

$$\hat{\lambda} = \frac{n_1}{p t}$$

where p is given in part (b).

- (d) $N_2(t)$ represents infections that have occurred by time t but are not yet detected. These are the complement of the thinned process, occurring at rate $\lambda(1-p)$. Using our estimate $\hat{\lambda}$:

$$\begin{aligned} E[N_2(t)] &= \hat{\lambda}(1-p)t \\ &= \frac{n_1}{pt} \cdot (1-p)t \\ &= n_1 \cdot \frac{1-p}{p} \end{aligned}$$

Substituting $p = 1 - \frac{1-e^{-\mu t}}{\mu t}$:

$$E[N_2(t)] = n_1 \cdot \frac{\frac{1-e^{-\mu t}}{\mu t}}{1 - \frac{1-e^{-\mu t}}{\mu t}}$$

Problem 4. Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ that is independent of the sequence $X_1, X_2, \dots$ of independent and identically distributed random variables with mean μ and variance σ^2 . Find

$$\text{Cov} \left(N(t), \sum_{i=1}^{N(t)} X_i \right)$$

Solution. We compute the following first,

$$\begin{aligned} E \left[\sum_{i=1}^{N(t)} X_i \right] &= E[\mu N(t)] = \mu \lambda t \\ E \left[N(t) \sum_{i=1}^{N(t)} X_i \right] &= E \left[E \left(N(t) \sum_{i=1}^{N(t)} X_i \mid N(t) \right) \right] \\ &= E[\mu N^2(t)] = \mu(\lambda t + \lambda^2 t^2). \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Cov} \left(N(t), \sum_{i=1}^{N(t)} X_i \right) &= E \left[N(t) \sum_{i=1}^{N(t)} X_i \right] - E[N(t)] E \left[\sum_{i=1}^{N(t)} X_i \right] \\ &= \mu(\lambda t + \lambda^2 t^2) - \lambda t(\mu \lambda t) \\ &= \mu \lambda t \end{aligned}$$

Problem 5. Emails arrive at a server according to a non-homogeneous Poisson process with intensity function

$$\lambda(t) = 5 + 2t, \quad t \geq 0$$

where t is measured in hours.

- What is the expected number of emails that arrive between $t = 1$ and $t = 4$?
- What is the probability that exactly 10 emails arrive between $t = 2$ and $t = 3$?
- Given that 50 emails arrived between $t = 0$ and $t = 5$, what is the probability that at least 20 of them arrived after $t = 3$?

Solution.

- (a) For a non-homogeneous Poisson process, the number of arrivals in $[t_1, t_2]$ is Poisson with mean

$$m(t_1, t_2) = \int_{t_1}^{t_2} \lambda(s) ds = \int_{t_1}^{t_2} (5 + 2s) ds$$

For $t_1 = 1, t_2 = 4$:

$$\begin{aligned} m(1, 4) &= \int_1^4 (5 + 2s) ds \\ &= [5s + s^2]_1^4 \\ &= (5(4) + 4^2) - (5(1) + 1^2) \\ &= (20 + 16) - (5 + 1) \\ &= 36 - 6 = 30 \end{aligned}$$

Expected number of emails = 30.

- (b) For $t_1 = 2, t_2 = 3$:

$$\begin{aligned} m(2, 3) &= \int_2^3 (5 + 2s) ds \\ &= [5s + s^2]_2^3 \\ &= (5(3) + 3^2) - (5(2) + 2^2) \\ &= (15 + 9) - (10 + 4) \\ &= 24 - 14 = 10 \end{aligned}$$

Thus $N(2, 3) \sim \text{Poisson}(10)$:

$$P(N(2, 3) = 10) = \frac{e^{-10} 10^{10}}{10!}$$

- (c) For $t_1 = 0, t_2 = 5$:

$$\begin{aligned} m(0, 5) &= \int_0^5 (5 + 2s) ds \\ &= [5s + s^2]_0^5 \\ &= (25 + 25) - 0 = 50 \end{aligned}$$

For arrivals after $t = 3$, i.e., $t \in [3, 5]$:

$$\begin{aligned} m(3, 5) &= \int_3^5 (5 + 2s) ds \\ &= [5s + s^2]_3^5 \\ &= (25 + 25) - (15 + 9) \\ &= 50 - 24 = 26 \end{aligned}$$

Given $N(0, 5) = 50$, the number of arrivals in $[3, 5]$ is binomial with parameters $n = 50$ and

$$p = \frac{m(3, 5)}{m(0, 5)} = \frac{26}{50} = 0.52$$

Thus:

$$P(\text{at least 20 in } [3, 5] \mid N(0, 5) = 50) = \sum_{k=20}^{50} \binom{50}{k} (0.52)^k (0.48)^{50-k}$$

Problem 6. An insurance company receives claims according to a Poisson process with rate $\lambda = 5$ claims per day. The claim amounts $X_1, X_2, \dots$ are independent and identically distributed random variables, independent of the claim arrival process, with mean $\mu = \$100$ and variance $\sigma^2 = \$50^2$. Let $S(t) = \sum_{i=1}^{N(t)} X_i$ be the total claim amount up to time t (in days).

- (a) Find the expected total claim amount $E[S(t)]$ and variance $\text{Var}(S(t))$.
- (b) What is the probability that no claims occur in a given day?
- (c) Given that 10 claims occurred in a day, what is the expected total claim amount for that day?
- (d) Suppose the company starts with \$10,000 at time $t = 0$ and earns interest at a continuous rate such that its capital at time t is $C(t) = 10000e^{0.05t} - S(t)$. Find the expected capital at time $t = 30$ days.

Solution. (a) For a compound Poisson process:

$$\begin{aligned} E[S(t)] &= E[N(t)] \cdot E[X] = \lambda t \cdot \mu \\ \text{Var}(S(t)) &= E[N(t)] \cdot \text{Var}(X) + \text{Var}(N(t)) \cdot (E[X])^2 \\ &= \lambda t \cdot \sigma^2 + \lambda t \cdot \mu^2 \\ &= \lambda t(\sigma^2 + \mu^2) \end{aligned}$$

With $\lambda = 5$, $\mu = 100$, $\sigma^2 = 2500$:

$$\begin{aligned} E[S(t)] &= 5t \cdot 100 = 500t \\ \text{Var}(S(t)) &= 5t(2500 + 10000) = 5t \cdot 12500 = 62500t \end{aligned}$$

(b) $P(N(1) = 0) = e^{-\lambda} = e^{-5} \approx 0.0067$

(c) Given $N(1) = 10$:

$$E[S(1) \mid N(1) = 10] = E\left[\sum_{i=1}^{10} X_i\right] = 10 \cdot \mu = 10 \cdot 100 = \$1000$$

(d)

$$\begin{aligned} E[C(30)] &= E[10000e^{0.05 \times 30} - S(30)] \\ &= 10000e^{1.5} - E[S(30)] \\ &= 10000e^{1.5} - 500 \times 30 \\ &= 10000e^{1.5} - 15000 \\ &\approx 10000 \times 4.4817 - 15000 \\ &\approx 44817 - 15000 = \$29,817 \end{aligned}$$

Problem 7. An insurance company models the number of claims $N(t)$ arriving by time t (in days) as a **mixed Poisson process** where the claim rate Λ is random. Suppose:

- Λ takes only two possible values:

$$\Lambda = \begin{cases} 2 \text{ claims/day} & \text{with probability } \frac{1}{2} \\ 6 \text{ claims/day} & \text{with probability } \frac{1}{2} \end{cases}$$

- Given $\Lambda = \lambda$, claims arrive as a Poisson process with rate λ .

- (a) Find $E[N(t)]$ and $\text{Var}(N(t))$.
- (b) What is $P(N(1) = 0)$, the probability of no claims in the first day?
- (c) Suppose we observe that $N(1) = 4$. What is the conditional probability that $\Lambda = 6$?
- (d) Using part (c), find $E[\Lambda \mid N(1) = 4]$.

Solution.

(a)

$$E[N(t)] = E[\Lambda] \cdot t$$

$$E[\Lambda] = \frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 6 = 1 + 3 = 4$$

So:

$$E[N(t)] = 4t$$

For variance:

$$\begin{aligned} \text{Var}(N(t)) &= E[\text{Var}(N(t) \mid \Lambda)] + \text{Var}(E[N(t) \mid \Lambda]) \\ &= E[\Lambda t] + \text{Var}(\Lambda t) \\ &= tE[\Lambda] + t^2 \text{Var}(\Lambda) \end{aligned}$$

First compute $E[\Lambda^2] = \frac{1}{2}(2^2) + \frac{1}{2}(6^2) = \frac{1}{2}(4) + \frac{1}{2}(36) = 2 + 18 = 20$ So:

$$\text{Var}(\Lambda) = E[\Lambda^2] - (E[\Lambda])^2 = 20 - 16 = 4$$

Thus:

$$\text{Var}(N(t)) = 4t + 4t^2$$

(b)

$$\begin{aligned} P(N(1) = 0) &= P(\Lambda = 2) \cdot P(N(1) = 0 \mid \Lambda = 2) + P(\Lambda = 6) \cdot P(N(1) = 0 \mid \Lambda = 6) \\ &= \frac{1}{2} \cdot e^{-2} + \frac{1}{2} \cdot e^{-6} \\ &= \frac{e^{-2} + e^{-6}}{2} \end{aligned}$$

(c) By Bayes:

$$P(\Lambda = 6 \mid N(1) = 4) = \frac{P(N(1) = 4 \mid \Lambda = 6) \cdot P(\Lambda = 6)}{P(N(1) = 4)}$$

First, compute $P(N(1) = 4 \mid \Lambda = \lambda) = \frac{e^{-\lambda} \lambda^4}{4!}$ So:

$$P(N(1) = 4 \mid \Lambda = 2) = \frac{e^{-2} \cdot 2^4}{24} = \frac{e^{-2} \cdot 16}{24} = \frac{2}{3}e^{-2}$$

$$P(N(1) = 4 \mid \Lambda = 6) = \frac{e^{-6} \cdot 6^4}{24} = \frac{e^{-6} \cdot 1296}{24} = 54e^{-6}$$

Thus:

$$P(N(1) = 4) = \frac{1}{2} \cdot \frac{2}{3}e^{-2} + \frac{1}{2} \cdot 54e^{-6} = \frac{1}{3}e^{-2} + 27e^{-6}$$

Now plug into Bayes:

$$P(\Lambda = 6 \mid N(1) = 4) = \frac{\frac{1}{2} \cdot 54e^{-6}}{\frac{1}{3}e^{-2} + 27e^{-6}}$$

$$= \frac{27e^{-6}}{\frac{1}{3}e^{-2} + 27e^{-6}}$$

Multiply numerator and denominator by $3e^6$:

$$= \frac{81}{e^4 + 81}$$

(d)

$$E[\Lambda \mid N(1) = 4] = 2 \cdot P(\Lambda = 2 \mid N(1) = 4) + 6 \cdot P(\Lambda = 6 \mid N(1) = 4)$$

From (c):

$$P(\Lambda = 6 \mid N(1) = 4) = \frac{81}{e^4 + 81}$$

$$P(\Lambda = 2 \mid N(1) = 4) = 1 - \frac{81}{e^4 + 81} = \frac{e^4}{e^4 + 81}$$

Thus:

$$\begin{aligned} E[\Lambda \mid N(1) = 4] &= 2 \cdot \frac{e^4}{e^4 + 81} + 6 \cdot \frac{81}{e^4 + 81} \\ &= \frac{2e^4 + 486}{e^4 + 81} \end{aligned}$$

If $e^4 \approx 54.6$, this is approximately:

$$\frac{109.2 + 486}{54.6 + 81} = \frac{595.2}{135.6} \approx 4.39$$